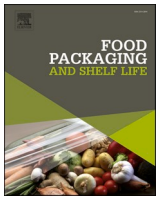
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Moulded pulp fibers for disposable food packaging: A state-of-the-art review

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ABSTRACT

The paper provides a comprehensive review of moulded pulp production, properties and advantages over single-use plastics and bioplastics for food packaging. Bagasse, wheat and bamboo are replacing wood as the fiber sources due to their cost effectiveness, wide availability and rapid renewability. Non-wood pulping and thermoforming processes have also advanced greatly which lead to improved fiber recovery, production efficiency and product quality. Challenges include lengthy production cycles, high chemical and energy uses, competition from bioplastics, and presence of toxic per-and poly-fluoroalkyl substances (PFAS). The industry is evolving rapidly to overcome these challenges including new laminated dry thermoforming processes, natural additives, customized fiber mixes, and PFAS-free products. Life-cycle assessments and provision of clear, viable green waste and recycling pathways relative to plastic alternatives are still needed to establish and maintain the competitiveness of cellulosic bioproducts with plastics. Critical research areas are proposed.

1. Introduction

Growth of plastics has outpaced almost all other manufactured materials; from 2 million metric tons (MT) produced per annum in 1950 to 322 million MT in 2015 (Geyer et al., 2017). Packaging creates the largest single demand for plastics and contribution to plastic waste (Fig. 1). Over 40%of all industrial packaging is devoted to food wrapping and packaging (a global market worth over $56 Bn) with 40% of food packages made from plastics (Rattanawongkun et al., 2020) (Fig. 2).

Non-recyclable/compostable packaging creates enormous demand on landfills (Berthet et al., 2016; pp. 8, 4252; Breslin, 1993) and significant environmental (land, river and ocean) pollution (Barnes et al., 2009; Jambeck et al., 2015). The scale of resource use and waste from the take-out food industry is enormous and growing (Molina-Besch, 2020). China — the world's largest producer and consumer of take-out food packaging — generates close to 350 kT/day of take-out packaging with 40 billion plastic take-out food boxes discarded in 2019 (Zhou et al., 2020). Food packaging accounts for about a third of China's total manufacturing and waste stream pollution and its water demand (Zhou et al., 2020). Canada generates 3 MT of plastic waste annually representing up to $8 billion per year in lost value in terms of wasted resources and energy (Government of Canada, 2020; Walji, 2020). Take-out containers make up a significant portion of this waste.

The plastic recycling rate in the EU is around 30% (Briard et al., 2018) but just 9% in Canada; significantly lower than the global average of about 15%. Impediments to effective plastic recycling include the large variety of polymers, co-additives, fillers, liners and reinforcements, colourants, and plasticizers (Andrady & Neal, 2009; Song et al., 2009). So-called 'greener' plastics such as oxo-degradable plastics containing co-additives to enable degradation under ultraviolet (UV) light and oxygen create further complications for waste management and contribute to litter by consumers thinking they easily biodegrade (Hann et al., 2017). Litter degradation can be incomplete and incineration or melting of oxo-degradable plastics can release small molecule biotoxins (such as brominated flame retardants, nano-additives), and require more complex separation of plastic waste (Das et al., 2021; Leslie et al., 2016). Before 2017 70% of plastic waste discarded by Western countries was

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exported to East Asia and the Pacific of which 89% was polymers commonly used in single use plastic food packaging; i.e. polyethylene, polypropylene and polyethylene terephthalate (PET). China discontinued these imports in 2017 followed later by others such as Malaysia; with an estimated 111 million MT of plastic waste estimated to be displaced by the Chinese policy alone by 2030 (Brooks et al., 2018).

Canada started progressively phasing out single use plastics in 2021 (Government of Canada, 2020) and China has launched various policies and incentives to transition towards re-usable/shareable or compostable/recyclable plant fiber packaging (Zhou et al., 2020). One option is moulded pulp; a relatively 'simple' and mature industry developed in the late 19th century for wood pulp as a pathway to recycle wood pulp, paper and cardboard. A wide range of disposable and biodegradable packaging materials, planter pots, food containers and trays were produced using the technology before the emergence of the plastics industry. The very first disposable food plates and trays were made from moulded pulp but replaced in the 1970's by petroleum plastics especially expanded polystyrene (XPS). These materials were promoted as cleaner, 'more-modern', cheaper, mass produced and highly process- and material-efficient. Environment pollution has led to restrictions on some of these, e.g., XPS, fostering a transition back to moulded pulp and newer 'biodegradable' bioplastics in the disposable food container market. While bioplastics are marketed as 'green including plant starch and polysaccharide-based degradable bioplastics; around half of all bioplastics produced are non-biodegradable (Rahman & Bhoi, 2021). Those that are biodegradable require tightly controlled temperature, moisture and bacteria/fungal composting conditions, are very variable in degradation rates, and some release eco-toxins, nano-particles and Green House Gases (GHGs) during decomposition (Adhikari et al., 2016; Emadian et al., 2017; Souza & Fernando, 2016). Bioplastics use starches and polysaccharides from edible parts of crops (e.g., corn kernels, potatoes, sugar beets) grown and transported specifically for purpose (Madival et al., 2009), whereas moulded pulp uses only inedible fibrous wastes (stalks, leaves, seed pods) from primary food crops including sugarcane, oil-palm, corn and wheat.

Reviews (Dey et al., 2020; Didone et al., 2017; Zhang et al., 2022) exist on moulded pulp products and manufacturing technology. However, they do not place the packaging option in the wider context nor focus on the specific manufacturing, additives and requirements of the take-out food container. The objectives of this paper are to provide a comprehensive review on the manufacturing, process and performance additives, and properties pertinent to food packaging using non-wood fibers. Challenges and research gaps in relation to this important and growing segment of the disposable food package industry are presented.

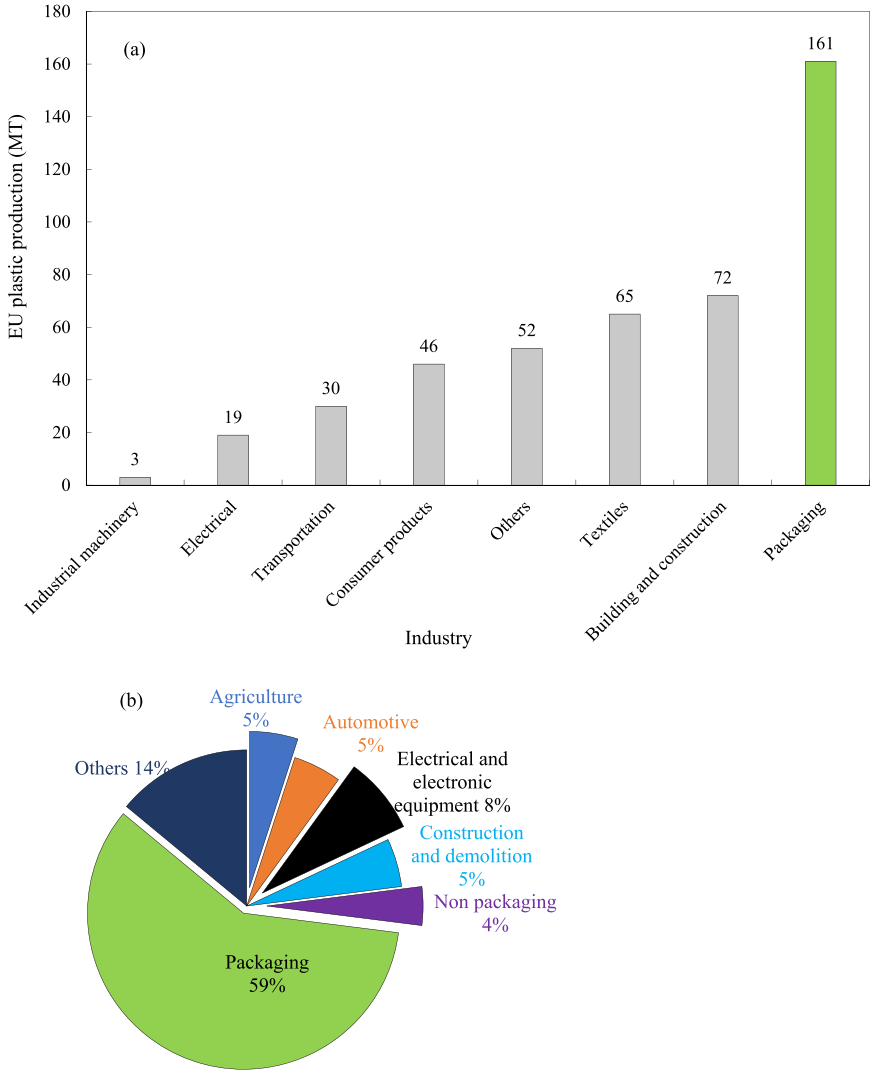


Fig. 1. Plastics production and waste (a) global production and use by sector, (b) EU plastic waste generation in 2015. [Adapted from: (a) Geyer et al. (2017) and (b) Tiseo (2018)].

2. Progress in moulded pulp food package manufacturing

2.1. Process overview and comparisons

Traditional moulded pulp production and newer processes for moulded pulp are comprehensively covered in various other publications (Dey et al., 2020; Didone & Tosello, 2019; Didone et al., 2017; Dislaire et al., 2021; Hogarth, 2005; Paine, 1990). The main process pathways are only briefly described and visually compared in Fig. 3 below with a focus on food container fabrication. Disposable take-out containers (e.g., plates, clamshells) use thermoforming processes for enhanced efficiency, smoothness, appearance and functionality.

1) *One-cast*: The original 'plain moulding' or 'thick-wall' process (Keyes, 1890, 1903) is where relatively smooth surface is vacuum formed with the forming wire mesh on one side and very rough surface on the open side, before the web is deposited onto a conveyor for drying. Wall thickness is typically 6 to 8 mm and the forming mould may be concave for heavy-duty machine parts or furniture packaging. These are commonly made from recycled paper and card.

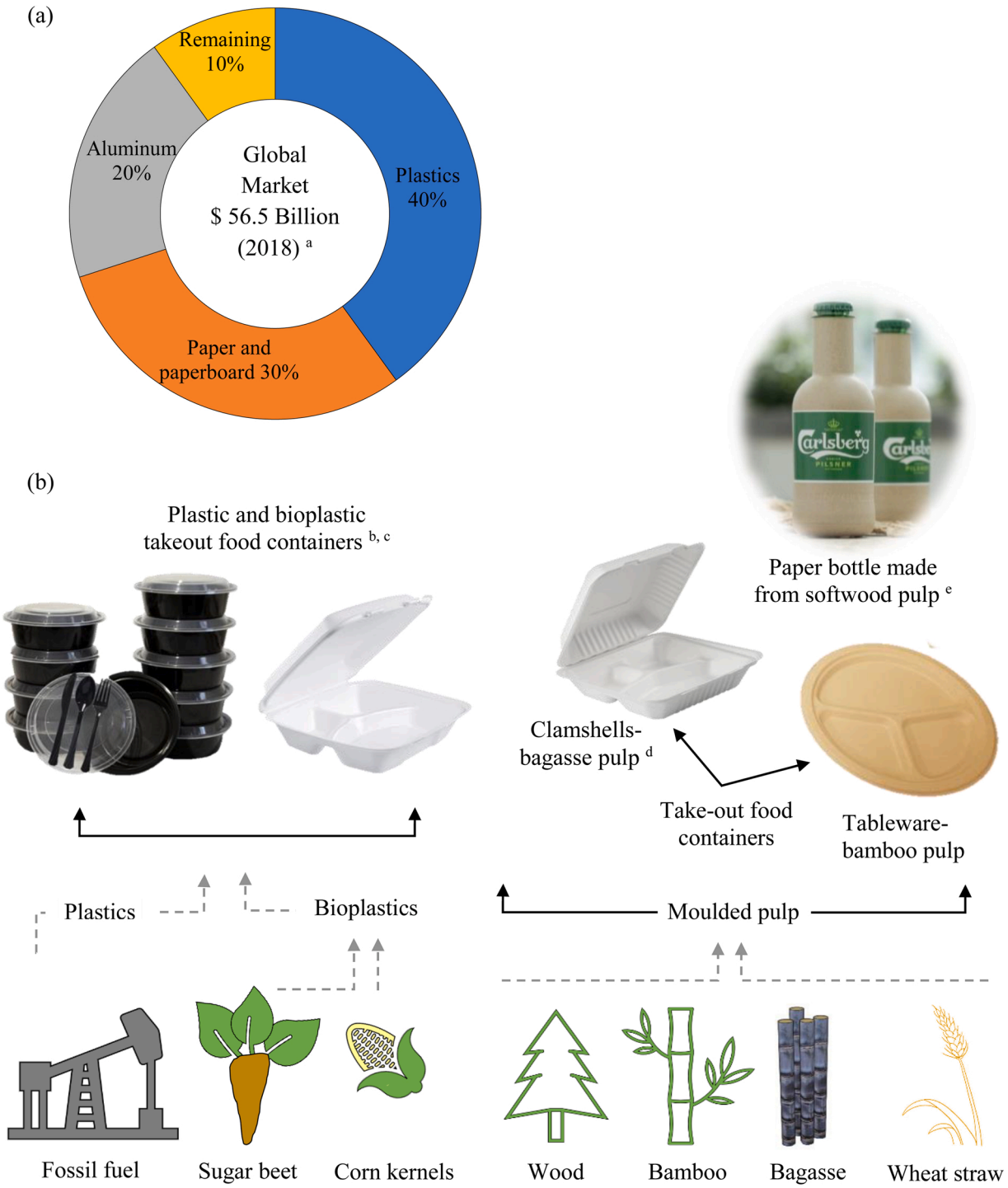


Fig. 2. Packaging sector (a) market by type and (b) material inputs for take-out food containers. [a] Grand View Research (2019), [b] ePackageSupply (2022), [c] Amazon (2022), [d] Uline (2022), [e] Paboco (2022).

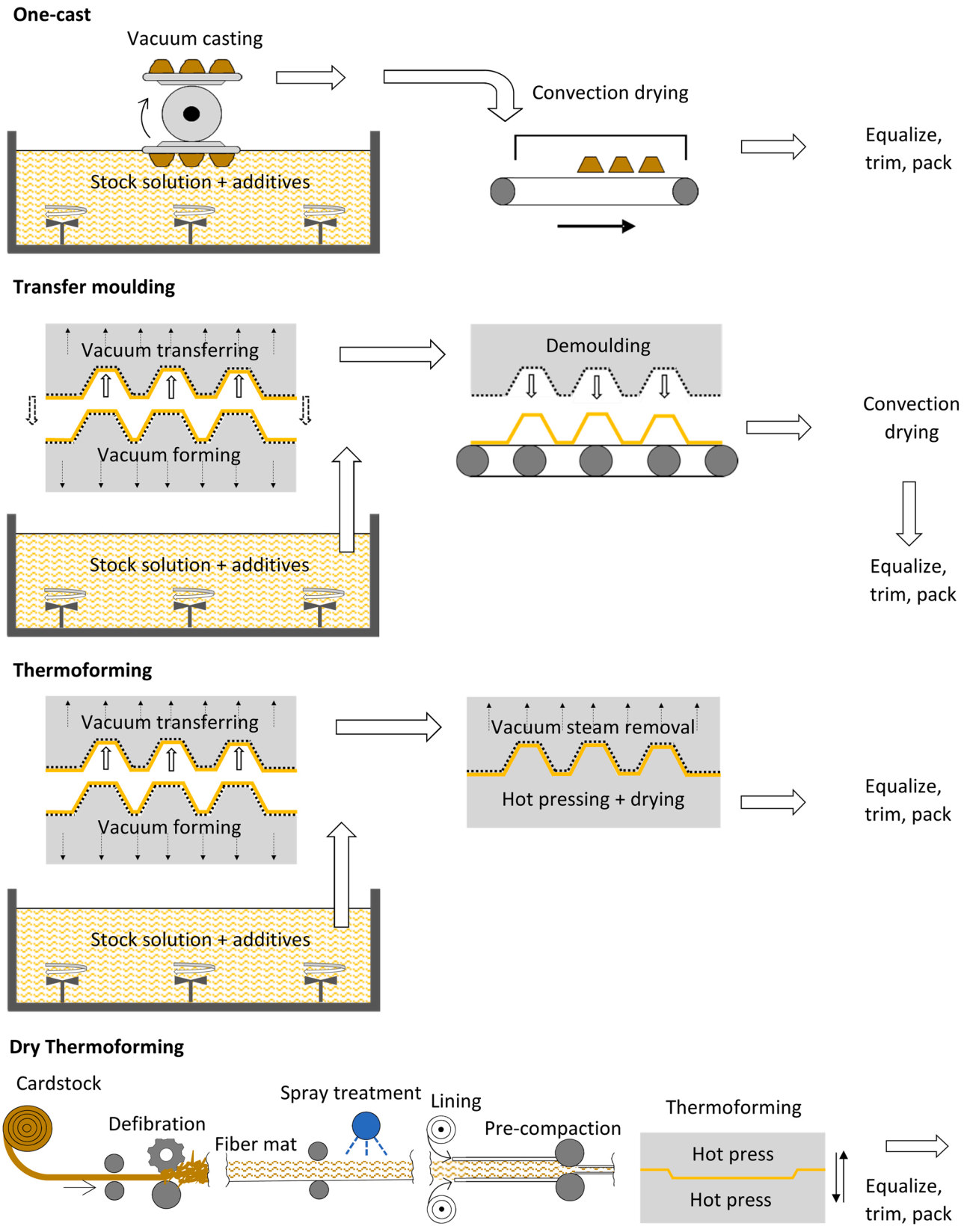


Fig. 3. Comparative schematic representations of moulded pulp processing methods: (1) One-cast, (2) Transfer moulding, (3) Thermoforming, and (4) Dry thermoforming.

2) *Transfer Moulding*: The cast is vacuum formed, dewatered and then removed from the forming mould to the transfer mould before being dropped off to a drying stage. The casts emerge with the wire imprint on the outside and a smoother inner surface than single casts. This process is high volume at > 20 tons/day (Didone et al., 2017), highly adaptable and produces a wide range of 'light-duty' moulded pulp packaging items with wall thickness typically ranging from 3-5 mm. It produces a wide range of electronic, household and hardware packaging such as computer protectors, egg cartons, cup holders, wine shippers and pulp urinals (Didone et al., 2017; Hogarth, 2005; Waldman, 2009). The dry web may also be transferred to a hot-press for further surface smoothening and embossing or logo printing.
3) *Thermoforming*: Also termed 'precision moulding' or 'thin-wall' processes, thermoforming eliminates convection drying of the cast to produce high precision, smooth, thin-walled casts whereby hot pressing, curing and drying take place in one step (Didone & Tosello, 2019; Didone et al., 2017; Dislaire et al., 2021). The process is faster and more process controlled but has higher production costs and energy consumption and is usually smaller in scale (Didone et al., 2017). It produces plastic-appearance packages from 0.4 to 1.5 mm in thickness (TRIDAS, 2022), and is commonly used in take-out food containers, smooth tableware, beauty product and high-end electronics packaging.
4) *Dry Thermoforming*: This is a new technology pathway for food and non-food packaging to adopt some of the process efficiencies of polymer/metal thermoforming and drastically cut water, drying time and energy requirements. Laminate thermoforming uses defibrated cardstock (Fig. 3 bottom) or hot-pressed paperboard laminates (Khakalo et al., 2017). Mechanically separated fibers are re-distributed into a thicker mat then sprayed with a food-grade cellulose-compatible biopolymer such as gelatin or guar gum, lined top and bottom with thin tissue and the 3-layer composite thermoformed to shape. Much developmental work has been done to optimize the stock material and processing parameters to commercialize this technology (Leminen et al., 2013; Leminen et al., 2020; Östlund et al., 2011; Tanninen et al., 2014; Tanninen et al., 2016; Tanninen et al., 2020; Vishtal & Retulainen, 2012). Whole paperboard may require mechanical bi-axial compaction stressing with rubber sheeting to increase sheet extensibility and reduce anisotropy of the fiber network during thermoforming which significantly improves the tensile strength, stiffness, temperature and humidity stability (Khakalo et al., 2017). Product thickness, network permeability/porosity, evenness of mat thickness and mechanical properties are better controlled by overlaying and thermoforming thin treated papers.

Further details on the hydropulping and vacuum forming and dewatering are summarized in Fig. 4. The hydropulping time and stock dilution must be controlled as it affects the consistency and morphology of the cellulose fibers and their fibrillation. Longer times create a higher water-retaining pulp with lower cellulose crystallinity (Chen et al., 2012; Dislaire et al., 2021). Prepared pulp stock is successively diluted to reach a consistency of < 1%, mostly 0.3% to 0.5% (Liu, 2020) and little research has been carried out on optimizing this process (Cho et al., 2009). Web porosity and permeability changes during vacuum forming and pressing and its drainage rate and moisture retention critically affect the economics of moulded pulp manufacture (Park et al., 2006).

Factors such as mesh hole size, stock solution pulping time, temperature, fines, contaminants and additives such as polyelectrolyte all influence the dewatering rate (Sung et al., 2004). Excessive fiber refining (beating) and hydropulping can damage the fine fibers needed for filling void space in the mat and interacting with the moisture and oil resistant additives necessary in food container manufacture. For example the effect of hot oil resistant additives is reduced by the extra alkali and beating rate required of bamboo pulp (Peng et al., 2007; Peng et al., 2006). Porosity and permeability also influence the thermal conductivity and barrier properties of the final product; both of which are critical for moulded pulp take-out food containers.

The drying schedule and time affect the consistency of wall thickness in moulded pulp and is critical to prevent deformation and friability of casts. The lengthy drying stage and recent changes in the fiber sources and types of oil repellant additives used in moulded pulp foodware manufacture (see Section 2.4.5) involve a transition to hot press thermoforming. Key parameters reported in precision moulding with a hot press step and thermoforming are summarized in Table 1.

Hot pressing takes advantage of the lower glass transition point for residual lignin and hemicelluloses in the pulp (Back & Salmen, 1982) and activates starch- and wax-based additives that help bond and consolidate the fiber network. Wang et al. (2021) developed fast-pressed, moisture resistant moulded pulp food containers from

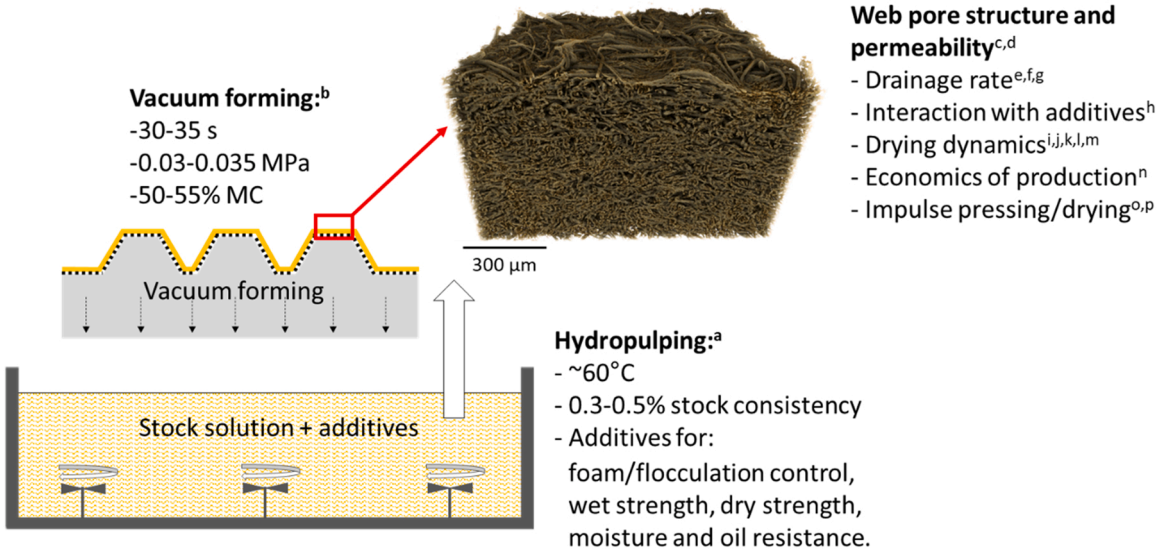


Fig. 4. Schematic and key parameters of hydropulping and vacuum forming stages: [a] C. Liu et al. (2020), [b]Hogarth (2005), [c]Bliesner (1963), [d]Nilsson and Stenström (1997), [e]Labrecque (1968), [f]Lindsay & Brady (1993a), [g]Lindsay & Brady (1993b), [h]Park et al. (2006), Couture, Jomaa, and Puigalli (1994), [j]Nilsson & Stenström (1995a), [k]Nilsson and Stenström (1997), [l]Nilsson et al. (1993), [m]Nilsson & Stenström (1995b), [n]Sung et al. (2004), [o]Didone et al. (2017), [p]Lucisano and Martinez (2001).

Table 1
Summary of hot-pressing parameters for moulded pulp.

Process	Cold press time (s)	Hot press time (s)	Hot press temperature (°C)	Reference
Precision moulding	–	NA	140-150 (15-25% MC)	Huang et al. (2009)
Thermoforming	25-30	90-120	185-190	Liu B.F (2020)
Thermoforming	–	45	220	Dislaire and Seantier (2021)
Dry thermoforming	–	5	160	Wang et al. (2021)

bamboo whereby slivers are pre-treated with warm nitric acid and hydrogen peroxide for 48 h then washed, refined into fiber, dried and hot pressed into casts at 160 °C at 0.5 MPa for 5 min. Residual lignin in the fiber acts as an adhesive and porosity reducer, imparting far superior strength and moisture resistance to a container that is still fully biodegradable and made without additives or liners. Higher moulding temperatures (220–250 °C) increase the crystallinity index of cellulose, reduces its hygroscopicity and produces a stronger and less porous network that is more resistant to moisture absorption (Dislaire et al., 2021). High performance structural moulded pulp is formed by hot pressing pure cellulose fiber (i.e., most lignin and hemicelluloses removed) that is more thermoplastic. The fiber is pre-treated with enzymatic hydrolysis lignin (EHL) as a binder whose polycondensation during hot pressing forms extra 'adhesive' bonds which contribute significantly to the strength and moisture resistance of moulded pulp (Faix, 1991; Sun et al., 2014; Wang et al., 2019; Wang et al., 2017). The compaction pressure of the pulp web is usually no more than 1 MPa and is set according to the raw pulp fiber bulk density and the required wall density of the final product (Dislaire et al., 2021).

2.2. Supply chain

A globally integrated supply chain is required to produce moulded pulp food packaging (Fig. 5). A food packaging company or supplier in North America or Europe usually contracts the package manufacture to an industrial partner usually located closer to the fiber source e.g. in China or Thailand. The order of suppliers starts at: 1) Grower (fiber supply), 2) Primary pulp mill (fiber raw pulp production), 3) Moulded pulp factory (package manufacture), 4) Packaging company wholesaler to primary vendor (restaurants/packaged goods companies), 5) Secondary retailers (e.g. supermarkets/online ordering companies), and 6) Consumer to waste pathway (recycling/composting). The packaging company is responsible for 3rd party product food safety/compost testing, certification and labelling for sale. Vendors and consumers may provide feedback to the packaging company that informs further product research, design and improvements. An example of the global shift in fiber and product supply in fast food packaging is a global fried chicken chain adopting home-compostable bamboo fiber buckets, gravy containers and bags for all its Canadian restaurants in 2021. The chain is also replacing all plastic cutlery with biodegradable fiber-based cutlery made from bamboo, corn and bagasse fiber. Canada is a global wood fiber producer yet fast food packaging is increasingly produced out-of-country with non-wood crop and bamboo fiber.

Before a product is commercialized and put into serial production the design (product, moulds and tools) and pilot testing of properties and performance is carried out by the packaging company. The packaging company also makes consumer-driven adjustments to products. Moulded pulp products and performance are defined by the raw pulp type and properties, pulp beating degree, additives, mould shape and web forming conditions, pressing and drying conditions and finishing methods (Sikora & Danielewicz, 2019). Relevant codes and standards govern the quality and safety of food packaging products, which are accredited through third-party testing contracted by the packaging company. Computer aided design (CAD) and rapid prototyping tools, and performance modelling have greatly reduced the historically lengthy 'concept – to – production' phase, allowing moulded pulp to become a competitive replacement for XPS foams across a wide range of custom packaging design applications (Ma et al., 2004; Song et al., 2001). Moulding dies and screens, and hot press dies can now be 3-D printed from hard wearing nylon polymer or metal allowing rapid prototyping and customization.

2.3. Fiber sources

Rapid renewability, cost, availability and advances in pulping and moulding technology developments for non-wood fibers including bagasse, wheat and bamboo are replacing wood fiber for moulded pulp food package production (Curling et al., 2017; Liu, 2020). Aspects

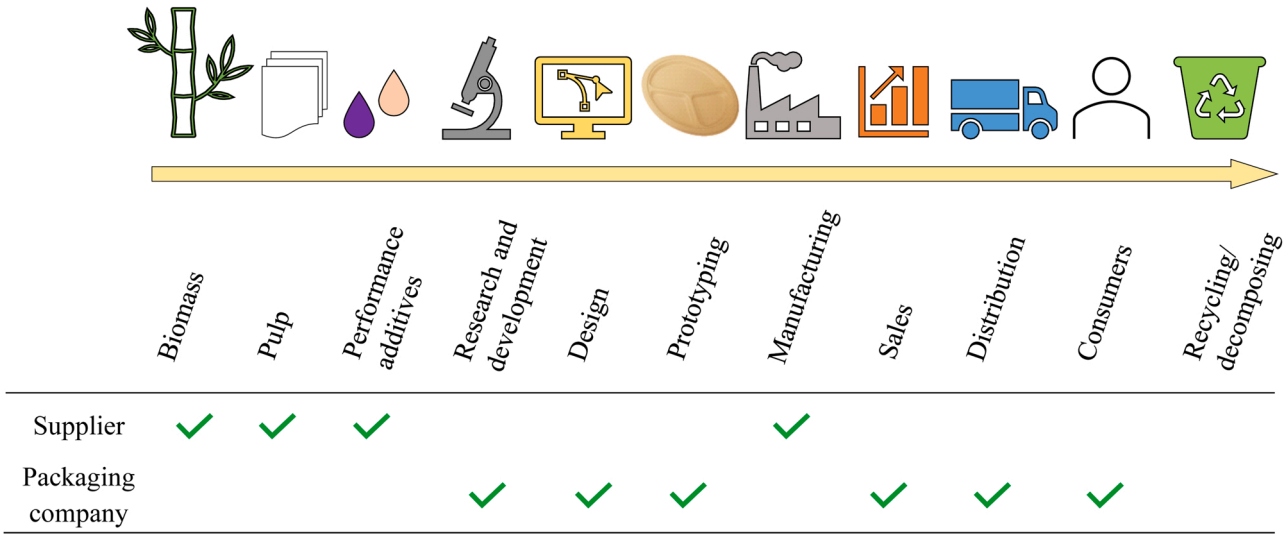


Fig. 5. Generalized supply chain for moulded pulp food packaging.

related to fiber for moulded pulp food packages are summarized in Table 2. TMP pulps are cost effective but have higher fines content producing a bulkier moulded pulp with greater moisture sensitivity (Sung, Kim, Kim, & Lee, 2015). Kraft wood fiber used in conventional paper and card manufacture is also not the optimal fiber type for manufacturing moulded pulp disposable food containers (Wang et al., 2021). Chemi-Thermomechanical pulp (CTMP) which may be bleached (BCTMP) is most common in moulded pulp production for its lower cost and higher fiber mass recovery (Dislaire, Grohens et al., 2021).

In North America, food safety regulations explicitly prohibit chlorine-bleached pulps in direct food contact packaging (Liu, 2018) but lack sufficiently detailed guidelines for the industry to confidently adopt recycled fiber for direct-contact food packaging (Bailey, 2015; Paquette, 1998; Song et al., 2000). In Europe, recycled fiber is permitted in food-contact packaging only under strict and clear regulatory guidelines (von Wright, 2006). Recovered fiber can contain a wide range of biological (fungi, bacteria) and non-food grade chemical (inks, dyes) contaminants posing food safety risks (Hurter, 2014; Luan et al., 2019). Non-fiber contaminants such as staples, glass and gravel disrupt production, and variable fiber sources and qualities also adversely impact the manufacture, appearance and performance of moulded pulp food packaging (Huo & Saito, 2009). Due to the cost and intensive processing required for moulded pulp (Liu B.F, 2020), alternative plant materials other than pulped fibers (e.g. leaves and powdered wastes) have been developed for disposable container market but these are outside the scope of this review.

2.4. Process and performance additives

Additives are necessary to meet a diverse range of processing and product quality/performance requirements. These are to increase the wet strength (mouldability) and dry bond strength of the fiber web, adjust pH to avoid flocculation of fines and other additives, increase web drainage, reduce air bubbles, improve product dimensional stability, moisture and oil resistance, improve ink printability, reduce fiber and production costs, improve whiteness, brightness and colour stability, and reduce energy consumption (Huang et al., 2009). A summary of the different types and functions of moulded pulp additives is given in Table 3.

Plant fibers are naturally hydrophilic and readily absorb moisture, grease and oils if not treated with additives used to give moulded pulp and food-grade papers their required moisture and grease resistance. Depending on fiber type and refining a complex interaction of hydrogen bonds, ionic bonds, dipolar interactions, induced polar interactions, van der Waals forces, and covalent forces (for wet strength resins) affect the strength and other properties of pulp and its products; both during formation and final product (Lindström et al., 2005). Additives can enhance bonding beyond those naturally occurring between fibers but their contribution is lower than the base level induced by fiber refining/beating to defibrillate (Lindström et al., 2005). The types and combinations of additives used in both paper and moulded pulp vary

Table 2
Fiber sources, pulp type and characteristics affecting moulded pulp food container production.

Pulp fiber source	Pulp type	Other substrates	Food safety requirements
Wood[a,d] Bagasse[p,s] Bamboo[p,t] Wheat[n] Corn[q] Oil palm[r] Blends[p]	Bleached chemi-thermomechanical pulp (BCTMP)[g] Non-chlorine bleached[f]	Whole leaves[o] Pulped leaves[p] Powdered lignocellulose[u] Binders: Activated starch[v] Polypropylene[v] Polyethylene[v]	Virgin fiber[b,c,n,h,i,j,k] No contaminants (inks/dyes, grit, metals, fungi/bacteria)[l,m] Chemical migration from substrate[b,e,h,w]

[a]Zhang et al. (2022), [b]Hurter (2014), [c]Hurter (2015), Kuiper et al. (2016), [e]Luan et al. (2019), [f]Liu Z (2018), [g](Dislaire and Seantier (2021), [h]Bailey (2015), [i]Paquette (1998), [j]Song et al. (2000), [k]FDA (2002), [l]Sung et al. (2015), [m]Huo & Saito (2009), [n]Curling et al. (2017), Dybka-Stępień et al. (2021), [p]B. F. Liu B.F (2020), [q]Jaroenpanyaprat (2012), [r]Majeed et al. (2013), [s]Aguilar-Rivera (2016), [t]H. Wang et al. (2021), [u]Xu (2008), [v]M. Zhou (2013), [w]Souza & Fernando (2016).

Table 3
Summary of additives used in moulded pulp processing and products.

Type		Example	Function
Process additives	Forming agent	Emulsion wax, anionic gelatin[a,g],	Increase set strength during forming, mat flexibility and conformability
	Retention/fixing/flocculation agent	Cationic polyacrylamide (PAM)[a]	Prevent clumping or loss of additives during drainage
	Antifoaming agent	Surfactant, e.g., fatty ester/amide, polyethylene glycol (PEG)[a]	Prevent voids and flaws during pressing/drying
	Wet strength agent	High MW cationic starch, e.g., potato starch[f]	Enhance fiber bond strength
	pH regulator	Aluminum sulphate[a]	Avoid flocculation
Properties/performance additives	Bonding/dry strength agent	High MW cationic starch, e.g., potato starch[b,c,d,e,f], plant proteins[i,j,k], CNF[l,m], enzymatic hydrolysis lignin (EHL)[n]	Enhance dry fiber bond strength
	Sizing wax/moisture repellant	Alkyl ketene dimer (AKD), alkyl succinic anhydride (ASA)[g,h], EHL[n]	Prevent moisture absorption and product softening, 're-pulping'
	Oil and grease repellant	PFAS[o,p], proprietary PFAS-free additives[q]	Prevent oil/grease permeating substrate causing failure or loss of insulating capacity
Mould control	Plant extract	*Listea cubeba* oil[r]	Reduce mould fungi in warm humid conditions
Barrier liners	Film-plastic/biopolymer Cross-linked biopolymers	Polyethylene terephthalate (PET)[s,t,u,v], chitosan[v,w,x], crosslinked cellulose[y] and hemicelluloses[z,a1], wey[b1,c1]/soybean[d1,e1] protein isolates, corn zein[f1,g1], wheat protein[h1,i1], poly(lactide) (PLA)[j1,k1],	Prevent liquid soaking of container substrate

[a]Sung et al. (2015), Hofreiter et al. (1981); [c]Lindström et al. (2005); [d]Marton (1991); [e]Hubbe & Gill (2016); [f]Huang et al. (2009); [g]Lindstrom & Soderberg (1986); [h]Sung et al. (2015); [i]Arboleda et al. (2014); [j]Jin et al. (2012); [k]Tayeb et al. (2017); [l]Brodin et al. (2014); [m]Hii et al. (2012); [n]Y. Zhao et al. (2020)); [o]Li & Liu (2003); Liu (2018); [q]Solenis (2022); [r]Suhem et al. (2019); [s]Frederiksen and Rasmussen (2003); Tanner et al. (2002); Tanner et al. (2002); Jeuniaux et al. (1988); Poulicek et al. (1986); [x]Haghighi et al. (2020); [y]Li et al. (2019); Anthony et al. (2015); Baldwin et al. (2012); [b1]Han & Krochta (1999); [c1]Han and Krochta (2001); [d1]H. J. Park et al. (2000); [e1]Rhim et al. (2006); [f1]Trezza and Vergano (1994); Trezza et al. (1998); [h1]Gällstedt & Hedenqvist (2006); Guillaume et al. (2010); [j1]J.-W. Rhim (2009); [k1]Rhim et al. (2007).

considerably and must be carefully tailored to the fiber source and product application requirements (Hogarth, 2005; Waldman, 2009).

2.4.1. Fiber bonding strength additives

Moulded pulp food container production uses additives common to paper manufacture including wet-end sizing waxes, cationic starches for wet and dry fiber bonding strength and synthetic resins to improve product strength and performance. Granular starches added to the pulp are cooked into matrix by heat and compression during drying and hot calendaring or pressing, contributing significantly to dry strength (Hubbe & Gill, 2016). In moulded pulp manufacture 1–2% cationic starch addition to the pulp slurry is necessary to prevent surface fiber dislodgement in the dried product and requires the transfer mould to be heated to 140–150 °C (Huang et al., 2009). Co-addition of cationic starch + engineered cellulose nanofibrils (CNF) (Brodin et al., 2014; Hii et al., 2012) or dialdehyde starch + chitosan (Korhonen et al., 2017) complements each other to produce a denser, stronger web when natural inter-fiber bonding is the limiting factor (e.g. recycled fiber, less beatable fiber). Plant proteins (such as soy) have also been developed as novel dry strength additives used in combination with starches; particularly for lignin-rich fibers and to counter sticky, hydrophobic pulp contaminants (Arboleda et al., 2014; Jin et al., 2012; Tayeb et al., 2017).

2.4.2. Foam control and wet moulding strength

Wet end additives are also used for foam and flocculation control and wet moulding strength of the fiber mat. Bubbles and flocculations greatly reduce web uniformity and wet forming strength leading to fissures at tension points and increasing its liquid and vapour absorbing capacity. These problems increase drying time and must be carefully controlled during manufacture (Sung et al., 2015). The antifoaming agent is generally a surfactant and may be oil, water or silicone-based. Foam and flocculation control agents include polyethylene glycol or alkyl polyacrylamide used to increase pulp drainability, reduce strain on pumps and equipment cleaning, and prevent defects (air bubbles, voids, breaks) in the moulded pulp from foam bubbles formed during pulp mixing and formation.

2.4.3. Moisture repellency

Sizing waxes such as alkyl ketene dimer (AKD), alkenyl succinic anhydride (ASA), or polyvinyl acrylamide (PVAm) are used to provide moisture repellency for the cellulose in pulp (Lindstrom & Soderberg, 1986) and are critical in the manufacture of all moulded pulp, especially food packaging. These additives enhance pulp web drainage and when used in higher amounts they help retain the structural and thermal insulation properties when the package is exposed to hot liquids and/or steam. Excessive moisture entering the substrate cleaves the natural hydrogen bonds between cellulose fibers and most moisture repellent agents involve fiber coating with a strong bond forming cross-linking agent that continues to bind fibers if the product becomes damp. The additives must bind to the fiber surface during pulp mixing and not be lost during de-watering. A cationic bonding or retention/fixing agent e.g. cationic polyacrylamide (PAM) stabilizer is commonly used. The fixing agent binds the water repellant to the pulp fibers reducing loss and drainage blocking during the vacuum dewatering process. Studies show additives become more concentrated in the outer layers of the container compared to the inner surface (Fassler, 2019).

Sung et al. (2015) provides data on water resistance of recycled wood fiber moulded pulps using various additive types including AKD, PVAm, and epoxy resin. AKD addition at 1% with 1% fixing agent (unspecified) was most effective at increasing moisture resistance, increasing average water surface contact angle from 12° on control samples to almost 90° in the case of moulded pulp made from recycled newsprint. Bleached Kraft pulp (BKP) commonly used in moulded pulp manufacture has a high bulk density and is very hydrophilic if used in raw form. This is easily rectified through addition of AKD at 1% to produce a hydrophobic surface and increase liquid contact angle to $> 100°$. AKD solid content in moulded pulp food containers is restricted to $\leq 0.4\%$ of total weight under FDA regulation 21 CFR 176.120 (FDA, 2022).

2.4.4. Oil and grease resistance

Unlined disposable hot food containers must contain a grease and oil resisting agent. Among the cheapest and most effective of these for moulded pulp and other paper/textile/fabric/carpet products are fluoro-carbons (FC) such as per-and poly-fluoroalkyl substances (PFAS). When added to the pulp stock the fibers interact with the strong cationic (+) surface charge of PFAS which bonds strongly with the negatively charged OH- groups on the cellulose fibers creating strong repellence to water and lipids (Li & Liu, 2003; Liu, 2018). Additives required for fabrication of unbleached bagasse fiber disposable tableware (Liu, 2018) included a fluoride-based oil resistant agent (usually PFAS) at 2% addition, water resistant agent (unspecified) at 5% and retention agent (polyacrylamide or PAM) at 0.02%, resulting in around 7% of pulp dry weight as additives The effects of the co-addition of water and oil repellant additives are synergistic (Liu, 2020).

Problems associated with PFAS are discussed in Section 4.4 and certain fiber combinations such as bamboo with bagasse (Liu et al., 2020) can help reduce or even eliminate the need for PFAS in the product. Other natural polymer additives like activated or enzymatic hydrolysis lignin (EHL) are also of interest. EHL addition during moulded pulp manufacture was effective at increasing the tensile strength to 20.3 MPa and surface water contact angle to 95.0° (Zhao et al., 2020). FTIR and TG analysis showed enhanced fiber bonding and surface hydrophobicity and SEM observations showed that the addition of EHL was effective at filling void-space between fibers. Additives based on activated plant starches and waxes are being developed and tested as PFAS replacements. Rather than strongly repelling lipids from the fibre surface, the PFAS alternatives restrict flow pathways through the wall and react with food oils to alter their flow characteristics to prevent penetration through the substrate.

2.4.5. Linings and mould control

For containers subjected to super-heated liquids and steam (such as tea/coffee cups, noodle bowls and oven/microwaveable TV-dinner trays) a thin film of heat and moisture proof plastic such as PET is applied but this significantly hampers the recycling and biodegradability of the product (Kottasova, 2016; Recyclecoach, 2019).

Much effort has gone into developing bio-based and biodegradable alternatives for the petroleum plastic additives, linings and coatings with many based on chitosan (Table 3) After purification and derivatization chitosan may be dissolved in water at pH 6.2 and solution-cast into barrier films (Chenite et al., 2001) and can be added in dissolved form to wet pulp sheets before pressing (Gällstedt & Hedenqvist, 2006). Chitosan is also an effective dry-strength additive; and can be better at improving mechanical properties (burst and tear) of hemicellulose-extracted bagasse pulp than cationic starch sizing (Hamzeh et al., 2013). Progress has been made in improving the strength, moisture and oxygen barrier properties of chitosan through chemical modification and blending with other biopolymers (Haghighi et al., 2020).

A wide range of other naturally sourced, non-toxic linings and films ('cross-linked biopolymers') have been developed (Table 3) and commercialized to improve the oxygen barrier, moisture barrier (water vapour permeance and liquid permeability) and oil resistance properties of food paper and pulp products (Khwaldia et al., 2010) and full biodegradation (Dumée, 2022; Triantafillopoulos & Koukoulas, 2020). For food contact applications anti-mould chemicals are not used in moulded pulp manufacture. However in areas of high relative humidity mould is a major problem during storage and transport of natural fiber-based packaging (Suhem et al., 2019). *Listea cubeba* oil and heat treatment was effective at improving the mould resistance of a food container product made from hot-pressed bamboo pulp and starch (Suhem et al., 2019).

3. Product properties and performance

3.1. General requirements

A single-use food container must be light-weight for material, waste and transport efficiency (Vartiainen et al., 2014), manufactured using a time, resource and cost-efficient process, thermally insulating and resistant to hot contents (i.e. retain its structural integrity) and tailored to be fully recyclable and/or biodegradable. Considering these properties moulded pulp is much better positioned over bioplastics. Industrial applications of biopolymers are currently restricted by poor oxygen/-water vapour barrier properties, thermal resistance and mechanical properties, and cost (Zhong et al., 2020). These are higher in density than petroleum plastic (~1.2 g/cm^3), have a relatively low softening temperature of 60 °C and are only thermally resistant to 70 °C (Pavlovskay et al., 2020). Moulded pulp is versatile ranging in density from about 0.35 to 1.2 g/cm^3 (Curling et al., 2017; Rattanawongkun et al., 2020; Wang et al., 2019) with better thermal insulation/resistance (see Fig. 6 storage modulus) properties than plastics. At ≈ 0.033 g/cm^3 only foamed polystyrene (XPS) is lower in density (Curling et al., 2017) with excellent insulation properties. Efforts are therefore underway to develop biodegradable foamed starch trays and containers to replace XPS, e.g., from PLA (Parker et al., 2011) or cassava starch (Brant et al., 2018).

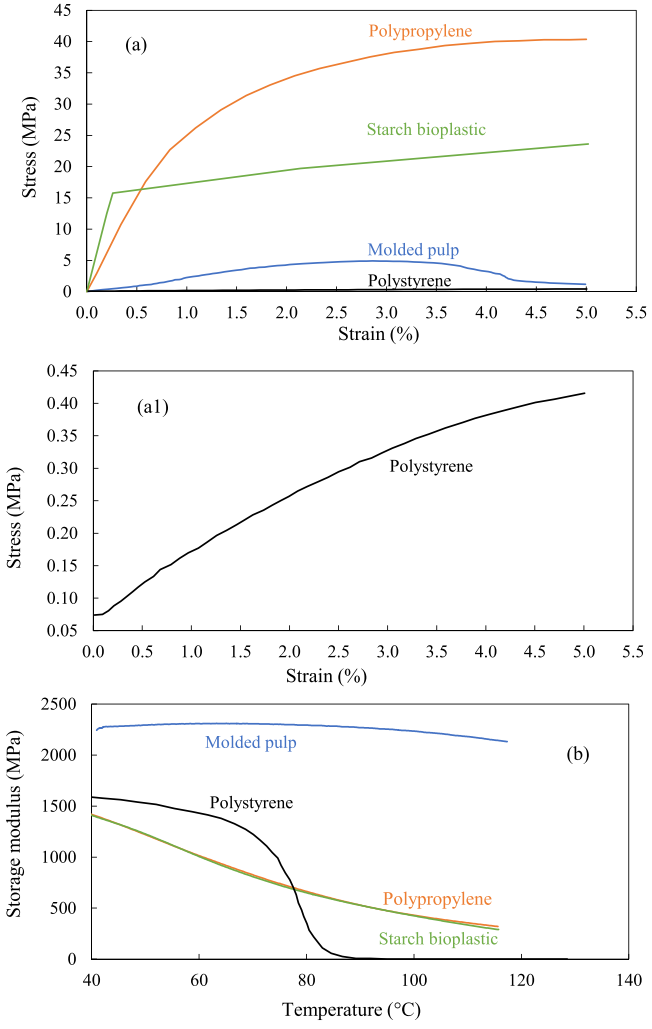


Fig. 6. Selected key mechanical properties of different materials in take-out food containers. a) stress-strain curves: polypropylene (Mohammadpour et al., 2014), starch bioplastic (Eterigho et al., 2018), moulded pulp (Gurav et al., 2003) and polystyrene (Chen et al., 2015); a1) magnification of polystyrene stress-strain curve; and b) storage modulus curves: moulded pulp, polypropylene, starch bioplastic (original data from authors), polystyrene (Ahmed et al., 2017).

The four basic material properties of packaging substrates that influence their quality and safety (Jayan et al., 2018; Kim et al., 2014; Chap. 2) are: 1) Mechanical properties, 2) Barrier properties, 3) Migration properties and 4) Chemical reactivity. The properties and performance of a moulded pulp cast is influenced by both the material properties (pulp fiber type and morphology, binders/additives, sheet density, flaws and voids, tensile and compressive moduli and strength, humidity effects) as well as the geometry and fabrication process of the form (Gurav et al., 2003; Hunt, 1998). The internal structure of moulded pulp products critically affects heat and mass transfer processes and underpins packaging durability, heat retention/loss and choice of additives/coatings (Didone & Tosello, 2019). Computed tomography (CT) is the established contactless technique for acquiring 3D images of the internal structure of soft, porous materials such as paper and moulded pulp products (Didone & Tosello, 2019; Saxena et al., 2017).

3.2. Mechanical properties

Moulded pulp has different mechanical properties to plastics used in disposable food containers. Comparative examples are shown in Fig. 6 for tensile stress-strain behaviour and storage modulus (loss of rigidity with heating). Moulded pulp being a light-weight, highly porous medium has low tensile stress absorbance compared with bioplastics or polypropylene but is higher than polystyrene (Figs. 6a and a1). However its non-plastic structure gives it very good retention of elastic modulus in flexure with heating (storage modulus) compared with polymer-based substrates (Fig. 6b) allowing it to retain rigidity in hot environments notwithstanding moisture absorbance.

Most physical and mechanical properties of moulded pulp are assessed using methods developed for paper and board products using TAPPI, ISO and in some cases British Standard or ASTM test methods (see Table 4).

Changes in flexural modulus with temperature and/or humidity may be assessed using Dynamic Mechanical Analysis (DMA) for dynamic modelling purposes. Since the mechanical properties of natural materials are strongly influenced by MC these are conditioned and tested under controlled, consistent environmental conditions (temperature and humidity) as stipulated in TAPPI Normative Standard T402 Sp-08 2013-Standard conditioning and testing atmospheres for paper, board, pulp hand sheets, and related products; to allow accurate comparison. Most properties tests should be conducted in a temperature and humidity-controlled climate chamber preferably at an RH of 50%. Information specific to the tests can be found in the relevant standards and the foregoing discussion summarizes selected properties and performance of non-wood fiber moulded pulp products.

Moulded pulp is significantly higher in tensile modulus (0.47 to

Table 4
Properties and testing standards for moulded pulp products.

Property	Test method
Thickness	TAPPI T411 om-97
Tensile	ISO 1994-2
Axial compression	TAPPI T 826
Flexural modulus	ASTM D4476
Crush/impact	ASTM D1709
Grease/oil resistance	TAPPI Test Method T 559 cm-12 Grease resistance test for paper and paperboard
Thermal conductivity	ISO 8302, GB 10297
Moisture Content	ISO 2017.287, TAPPI, T. 2004. 412 om-02
Moisture absorbance	TAPPI/ANSI Test Method T 441 om-13, ISO 2014.535
Vapor transfer	ASTM E96-95
Food migration	EN 1186
Biodegradability/ composability	ASTM D 5338, ASTM D6400, EN 13432, EN 14995 ISO14855, ISO 17088, BS ENV807

0.68 MPa) compared with 0.16 MPa for XPS (Curling et al., 2017) based on 20–40% wood fiber mixed with wheat pulp. However its bending modulus is lower (0.015 MPa) than that of XPS (0.035 MPa). The ultimate strength and elastic modulus of moulded pulp increase as the substrate density increases due to greater fiber-to-fiber contact (Alava & Niskanen, 2006; Ji & Wang, 2011) but this must be balanced against materials and transport costs, and thermal insulation capacity.

Some comparative data on material density and tensile strength are given in Table 5 for moulded pulp including more recently-developed high performance moulded pulps (Wang et al., 2019, 2021). Conventional moulded pulps used in take-out food containers can range from about 0.12 g/cm^3 to 0.8 g/cm^3 in density, with tensile strength ranging from about 0.5 to 4.5 MPa for wood moulded pulp substrates. Tensile strength values are higher than that of XPS (0.4 MPa). Moulded pulp is more material efficient in properties such as tensile strength and storage modulus (Fig. 6) than bioplastics PLA and cornstarch plastic substrates. These have higher density than most petroleum plastics for the same tensile strength range. High performance structural moulded pulp (SNFM) emerges in the density range of petroleum plastics but with a higher range of tensile strength (up to 17 MPa) demonstrating that emerging technologies for high performance moulded pulp can extend into many other applications of structural plastics.

Moulded pulp has unique and complex non-linear stress-strain behaviour (Ji & Wang, 2011; Ji et al., 2008; Wang et al., 2012). Its strength and stiffness parameters are more critical for the design and static compression modelling of the cushioning, shock absorbing and crush/collapse resistance properties (Gurav et al., 2003; Wang et al., 2012). However the crush/collapse resistance is key to food container performance particularly during shipping, storage and under high heat and moisture conditions (Wang et al., 2021). Didone et al. (2017) show that the structure of moulded pulp is less rigid (i.e., deforms more and fails earlier under lower tensile strain levels) than that of oriented paper sheets due to the random orientation of the fibers. The tensile stress-strain behaviour of moulded pulp includes not only elastic-plastic deformation, but also shows viscous, or thermoplastic, characteristics (Ji et al., 2008). The strain accumulation and distribution are uneven due to the porous, heterogeneous structure of voids and random orientation and size/shape fractioning of fibers.

Material density and the loading rate influence the wall compressive stress-strain relationships of moulded pulp (Ji & Wang, 2011). Increased density increases the elastic modulus and ultimate strength and the loading rate decreases elastic modulus and increases ultimate strength

Table 5
Comparative density and tensile strength for conventional and high-performance moulded pulp and other petroleum and bioplastics.

Material	Density (g/cm^3)	Tensile strength (MPa)	Reference
Expanded polystyrene (XPS)	0.028	0.38–0.42	Chen et al. (2015)
Moulded pulp–wheat straw	0.12–0.15	0.5–1.3	Wang et al. (2019)
Moulded pulp–wood	0.35	4.1–4.4	
Moulded pulp–wood (residual lignin pulp)	0.76	43.7	Wang et al. (2021)
Self-bonded natural fiber material (SNFM)	0.86–1.06	21.5–83.5	Wang et al. (2019)
Polypropylene (PP)	0.89–0.91	17.9–30.0	
High density polyethylene (HDPE)	0.94–0.95	17.0–20.8	
Wheat/beet starch plastic	1.17	15–19	Pavlovskay et al. (2020)
Polylactic acid (PLA)	1.24	10–33	Sepet et al. (2015), Subramaniam et al. (2019)
Cornstarch plastic	1.45	15–49	Spence et al. (1995)

estimates. Finite element models for the crushing behavior of different shapes of moulded pulp packaging have been developed using a Bilinear Kinematics Hardening model and the empirically determined parameters: Young's modulus, yield limit, shear modulus and Poisson's ratio (Wang et al., 2012). No data have been found specifically relating to deformation/collapse resistance of moulded pulp food containers under high heat and moisture. Data exist for wall material tensile strength and modulus but reports of physical tests (e.g. crushing or rigidity) on containers are rare. A purpose-designed apparatus (Tanninen et al., 2021) is used to test the rigidity of paper food containers using a three dimensional torsional apparatus. Tests showed that in order to maximize container stiffness the long edge of the container or tray should be cut to follow the machine direction of the pulp.

3.3. Barrier properties

3.3.1. Moisture and grease resistance

Transfer dynamics for moisture (liquid and vapour forms) as well as fats and oils from food are among the most critical functionalities of the moulded pulp single use food container. *Liquid water absorbance* of pulp sheets is measured based on the Cobb test for paper given in TAPPI/ANSI Test Method T 441 om-13 and ISO 2014.535 (Curling et al., 2017; Dey et al., 2020; Jaroenpanyaprat, 2012; Kimet al., 2015). Business paper generally has a Cobb value of 22–26 g/m^2, unsized paper $>$ 50 g/m^2, while typical values for bagasse moulded pulp tableware containing 5% moisture repellant are between 15–20 g/m^2 (Liu Z, 2018).

Water vapour uptake is the % increase in mass of water absorbed by the fiber substrate when exposed to a high humidity environment for a given time usually between 2 and 24 h. The *water vapour transmission rate* (WVTR) is the amount of water vapor (g) that can permeate through a given unit area (m^2) of packaging material over time (h) (Siracusa et al., 2008) and this value should be minimal (Han & Krochta, 1999). This is tested using a gravimetric method (Germadios et al., 1994) via ASTM E96–00 Standard test method for water vapour transmission of materials. WVTR is closely negatively correlated with substrate density (porosity) and largely independent of RH (Nilsson et al., 1993).

The effects of absorbed moisture on strength properties and performance of cardboard and moulded pulp packaging are drastic showing a 5–10% decrease in strength for every 1% increase in moisture (Marcondes, 1992; Sørensen & Hoffmann, 2003). Water vapour absorbance depends largely on the cellulosic fiber source and wet end additives such as moisture repellant sizing agents (Rhim & Lee, 2009). Few studies exist establishing the sorption isotherm – property relationships for paper-based packaging (Rhim & Lee, 2009). Water vapour sorption isotherms at 25 °C, 40°C and 50 °C were derived for common paper-based packaging substrates including Kraft liner (paper grocery bags etc.), machine glazed vegetable parchment (burger/sandwich wrappers), and solid bleached sulphate paper board (food and drink cups and cartons). Foss et al. (2003) show heat and mass transfer models for paper during moisture sorption from humid air and sorption isotherms. Compressive strength models have been established for moulded pulp food cups (Sørensen & Hoffmann, 2003) and paperboard (Parkeret al., 2006).

The take-out food container must also not transfer hot oils to the outside or lose its structural performance if it absorbs oil. For measuring oil/grease resistance, the test assays the spread outwards of droplets of various mix ratios of reference permeants castor oil, toluene, and n-heptane. Resistance levels range from 1 (lowest) to 12 (highest), and moulded pulp containers should ideally sit between 5 and 8 (Liu, 2018; Yang & Bao, 2002). For products made with non-PFAS grease/oil resistant additives a Harmonized Hot Oil Test standard for foodservice products (Foodservice Packaging Institute, 2013) is used. To test and qualify substrates a Cobb ring clamp is used to contain a 3-mm-deep layer of corn oil at 65 °C for 20 min on the sheet food contact surface. There must be no observed oil penetration stains on the bottom surface.

3.3.2. Thermal insulation

Thermal insulation is necessary to keep food and liquids hot until consumption. Thermal conductivity (λ) increases with increasing material density, MC, and ambient temperature. Functionality of moulded pulp food packaging such as heat and mass transfer and structural performance (mechano-sorptive behaviour and compressional creep) in changing temperature and humidity conditions are critical and have been modelled (Sørensen & Hoffmann, 2003, 2004; Tanner et al., 2002). Surprisingly there appears to be no readily-available standard for measuring thermal conductivity of thin moulded pulp food packaging substrates, or any (publicly available) data for λ of food containers. Most published data of the heat retention capacity of food and liquid containers/cups comes from literature on paediatric burns and scald admissions to hospitals from disposable cups and noodle packs (Greenhalgh et al., 2006). These studies measured and compared the cooling rates of hot test liquids and foods themselves using an inserted temperature thermocouple (Abraham et al., 2016; Jamnadas-Khoda et al., 2010; Naik et al, 2019). This highlights the need to balance insulating capacity (ability to keep contents hot) and safety (heat retention/burns risks) (Greenhalgh et al., 2006; Shalom et al., 2007).

Data from direct material testing of thermal conductivity of pulp based materials exist for pulp fiber insulation building materials (e.g. ISO 8032) to assess thick hollow core cardboard-based insulation panels (Čekon et al., 2017). Liu et al. (2012) used the hot wire method to measure thermal conductivity of thick steam-injection pressed pulp-resin thermal and sound insulating panels (80–100 kg/m^3 density) under Chinese Standard GB 10297. Wood and rice stalk fiber boards of 95 kg/m^3 density had thermal conductivity readings of 0.061 and 0.058 W/m^{-1} K^{-1}, respectively. The density of the abovementioned fiberboard products is around 30% of the density of the moulded pulp substrate tested by Gurav et al. (2003) and therefore likely have lower thermal conductivity than moulded pulp food containers. XPS insulation has even lower thermal conductivity, with values ranging from 0.035 to 0.045 W/m^{-1} K^{-1}.

3.4. Chemical reactivity and food migration

With increasingly complex food packaging materials, composites and liners, migration of toxic chemicals or nano-sized particles such as fillers and additives into food is a key safety and performance property (Arvanitoyannis & Bosnea, 2004; Castle, 2007; Paquette, 1998; Song et al., 2000). Migration risk is related to the physico-chemical properties of the migrating substance, the packaging material, and the food (e.g. acid or fat content), the temperature of the food and surrounding environment, storage time and package size in proportion to foods volume (Muncke, 2013).

Migration limits for food contact materials (FCMs) are covered under a harmonized legal EU framework, Regulation (EC) No 1935/2004 for all FCMs and include both Overall Migration limits of all possible compounds (OML) and Specific Migration limits (SML) for individual, mostly high-risk substances. Test procedures are given in the EN 1186 series using standardized food simulants of consistent composition and purity that imitate food behavior and possible chemical migration into the container substrate (Souza & Fernando, 2016).

While toxins like bisphenol A from polycarbonate plastics are well known (Vilarinho et al., 2019) other additives like nanoclay particles used in nanocomposite packaging are also of concern (Echegoyen et al., 2016; Muncke et al., 2014). Most pulp food containers are not printed as large pore sizes in the substrate can allow printing ink solvents through from the outside to the food-contact surface (Dima et al., 2011; Richter et al., 2009). Perfluoro additives - perfluorooctane sulfonate (PFOS) and perfluorooctanoic acid (PFOA) contaminants in paper coatings for moisture and oil resistance are bio-toxic and bio-persistent but difficult to trace using conventional GC/MS techniques (Begley et al., 2005). These have been found to transfer in detectable amounts (mg/kg levels) from the liner into food oils (Begley et al., 2005). A moulded pulp article must therefore be inert to migration into a much wider range of food stuffs than a 'for single specific contents' package. Its additives must therefore be as simple, non-toxic and few in variety as possible to achieve required functionality.

3.5. Biodegradability

The substrate must not degrade during shipping or storage, or contact with acids and oils in food but must be sufficiently reactive to break down after use in composting conditions, i.e. undergo oxidative, hydrolytic and enzymatic degradation. Under ISO/DIS 17088 packages must conform to four composting criteria: (1) Biodegradation (breakdown to water, CO_2, cellular biomass by natural microbes-mainly bacteria and fungi), (2) Disintegration, (3) No negative impacts on the biological processes, and (4) No toxic substances in final compost (toxic metals and non-degradable polymers) (Muniyasamy et al., 2013). For newly developed packaging materials or linings, meeting legislated biodegradation benchmarks takes precedence over mechanical and barrier properties (Siracusa et al., 2008). Traditional moulded pulp food packages such as berry or egg cartons are fully biodegradable, however some containers can contain a more complex range of moisture and grease resistant additives and sometimes co-polymers and plastic linings which can affect its ability to degrade rapidly and safely (i.e., not release biotoxins) in food compositing facilities. High lignin content in lignocellulosic fibers can also affect their degradation rate during composting (Muniyasamy et al., 2013).

Good reviews and listings (Briassoulis et al., 2010; De Wilde, 2020; Muniyasamy et al., 2013) cover International and National norms and standards relating to the biodegradability of substrates used in packaging and other applications. In North America it is strongly recommended that dispensers of take-out food containers (e.g., takeaway outlets, restaurants, food trucks) and municipal green waste management authorities only handle 'compostable' packaging that has been third party (e.g., Biodegradable Products Institute) certified to meet ASTM D6400 or ASTM D6868 standard series. These standards and ISO 17088–13 and ISO 14855–18, and EN 13432–2000 and EN 14995 cover specifications and tests for biodegradable packaging with emphasis on assessing 'compostable' bioplastics. The testing media and conditions pertain to certifying the material suitability for industrial composting facilities rather than home or garden composting bins (Briassoulis et al., 2010). Further information is available (Zhong et al., 2020) on bioplastics and biodegradable plastics, their constitution and biodegradation pathways and evaluation.

Most standards contain procedures for assessing biodegradability based on weight loss, CO_2 generation, change in tensile strength, residue dimensions and mass fractions, and chemical and physical properties (Singh & Sharma, 2008). A soil burial method maintaining moisture level at 20–40% through sprinkling, continuous or periodic monitoring of CO_2 generation and monitoring residual mass every 15 days is most common, but procedures are also available for measuring degradation rates in sea water and river mud environments (Jayan et al., 2018). Studies on biodegradability of bioplastics and films typically use EN, ASTM or ISO standard CO_2 generation and monitoring criteria in purpose-designed, oxygen-supplied soil jar tests (Aryal, 2019; Mazzon et al., 2022; Pischedda et al., 2019; Šerá et al., 2020), whereas the very few studies assessing biodegradation of moulded pulp containers (C. Liu C, 2020; Peng et al., 2007) use modified unsterile soil box methods with only visual degradation and mass loss criteria. Liu (2020) buried complete containers in natural soil for 30 days with visual and mass loss assessments at 15, 30, 45 and 60 days. In a study on straw-based moulded pulp (Curling et al., 2017) samples from the test products and placed into a soil box where the samples rested and were shallowly covered by unsterile soil kept moist at 30–35% MC at room temperature for 4 weeks. The method developed and used (Curling et al., 2017) was a non-standard test (Curling et al., 2002) based on the British standard method ENV807 (CEN, 2001). The criteria in the EN standards include a

biodegradation rate of 90% (relative to reference cellulose) within six months of industrial composting conditions at 58 °C. Within 12 weeks there must be no more than 10% of the original dry weight of the material retained on a 2 mm fraction sieve.

4. Challenges and outlook

4.1. Policy and market dynamics

Evolving government legislation and recent product evolution, PFAS-alternatives and adaptive manufacturing processes point to expanded opportunities for plant fibers and plant polymer plastics for single use food packaging. PFAS-free moulded pulp food packaging is a developing market requiring innovation in product manufacturing and design (Footprint, 2021). The market and regulatory environment for packaging materials is complex and often counterintuitive. Some major global food companies that use enormous quantities and varieties of food packaging disfavour compostable and plant-polymer derived materials (see Table 6 as an example from Britain). Common petroleum plastics like polypropylene, polyethylene and PET are on the Green-Preferred list while plant polymer plastics (PLA) and compostable and cellulose/regenerated cellulose materials are relegated to the Red-Exit or Amber-Hold and Seek-to-Replace lists. Another global food distributor also lists regenerated cellulose (e.g., cellophane commonly used in transparent package windows or wraps) as 'phase-out' materials for which recycling pathways are difficult and unlikely to be established. Others on that list include PVC and related polymers, polystyrene and XPS, and plastic-laminated paper products such folded packs and coffee cups.

This is based on various contributing factors such as poorly developed collection and sorting infrastructure for green waste/recyclables in some western countries including the UK (Poole, 2020). Petroleum plastics currently benefit from more established wash-and-reuse/recyclability/recycled-content pathways than bio-based materials. Polymers and PFASs in food-grade papers (e.g. moulded pulp/laminated card take-out packages, baking parchment, burger wraps) precludes them from green-waste municipal composting schemes, diverting them to landfill. Problems with this include methane produced by compostable packaging ending up in landfills and food/additive contaminants disrupting paper recycling streams (Poole, 2020).

The above policies assume efficient and adequately high recycling rates for petroleum plastics but even in industrialized countries like Canada or the UK plastic recycling rates remain low. Non-or partially degradable plastic garbage remain significant challenges in all countries. Such market signals create more challenges for plant fiber and other biopolymer-based food container materials to replace single use plastics. Further complicating the transition to sustainable packaging materials are the unclear definitions for plastics informing Europe's Single Use Plastics Directive (SUPD) (European Parliament, 2019). The Directive lacks clarity on relative material sustainability, LCA and biodegradation pathways for food packaging materials. A submission to SUPD (FinnCERES, 2021) strongly recommends the European Commission recognize and establish a reliable definition /category for benign bio-based and/or biodegradable man-made polymeric materials. With the emergence of PFAS-free moulded pulp products there is an urgent need for better fiber-based foodware compost/recycling streams allowing for balanced functionality and composting or recyclability.

Table 6
Categories for preferred or phase-out packaging materials for a major food and grocery distributor in the UK [Adapted from Poole (2020)].

Red–Exit *Poor for recycling and/or potentially harmful*	Amber–Hold *Until infrastructure and/or scientific development take place*	Green–Preferred *Easily recycled* *Can have high recycled content*
PVC and polystyrene	Home compostable E.g., Cellulose, Natureflex™ (Futamura, 2021)	Sustainably sourced wood, board, paper, and Glassine
Compostable /biodegradable materials	OPP–Oriented polypropylene	PET–Polyethylene terephthalate
Water-soluble plastics	Black plastic	Glass
PLA–Polylactic acid	PP–Polypropylene (*for certain food applications*)	PP–Polypropylene (non-food)
Oxy/oxo-degradable	Complex laminates	HDPE and LDPE
Polycarbonate	PVdC (Not PVC)	PE–Polyethylene Preferred material for flexible film)
Acrylic	New materials	Steel and Aluminium

4.2. Life cycle analysis

There is a dearth of reliable life cycle analysis (LCA) information and comparisons for bioplastic and fiber-based food packages. Results from a lifecycle assessment of disposable take-out food containers (Gallego-Schmid et al., 2019) indicate the raw materials extraction and processing, material efficiency (mass needed per functional unit), container manufacturing process and end of life (landfill, recycling, composting) contribute most to their ecological impacts. XPS was shown to be the least ecologically damaging package option in terms of material requirements per unit, processing and energy efficiency. The single-use disposable polypropylene container was ranked lowest in 7 of the 12 sustainability indicators but remains a staple of the take-out food container market. This alone creates a significant driver for the development of bio-degradable XPS products - a significant market challenger for moulded pulp in the disposable food container market.

Moulded pulp faces competition from an evolving and expanding range of bio-degradable disposable food containers made from thermoplastic plant starches (Khan et a., 2016). Degradable bioplastics (mainly starch-based and PLA) are the fastest growing segment of the so-called 'green packaging' market, valued at $US3.7 billion in 2019 with projected CAGR of 9.5% through to 2027 (Grand View Research, 2019). These are also problematic in terms of recycling/green waste pathways (Table 6). The Gallego-Schmid study did not include bioplastic or plant fiber based moulded pulp container or the growing subsector of wash-and-reuse cup and container schemes (Mugshare, 2020; Walji, 2020). Another LCA (Madival et al., 2009) compared PLA bioplastic clamshell containers with PET and XPS equivalents incorporating the production and transport of inputs (water, energy, fertilizer, pesticides, herbicides) required to grow the corn specifically to produce PLA. Their study found the weight of the container substrate and the combined transport phases for the different inputs and finished products to be the most significant contributors to the environmental impact of packaging systems. PET plastic was ranked highest in environmental impact. A global PLA monomer supply shortage also precludes this material from adequately meeting market needs for bio-based packaging (Jem & Tan, 2020).

4.3. Resource consumption in manufacturing

While the above considerations are Western-focused the quest for cost-effective, sustainable disposable packaging options in its largest market is even more challenging as some provinces have banned non-degradable single use plastics (Hou & Ma, 2019). Liu (2020) considered the viability of four classes of domestically produced disposable 'degradable' tableware on the market in China. These were: (1) 'degradable' polypropylene, (2) moulded pulp, (3) modified starch and (4) vegetable fiber. Their study noted problems with the practical biodegradability of 'biodegradable' polypropylene (PP) products which contain photosensitive additives to accelerate degradation under UV light. Modified starch containers were made from either thermoplastic

(gelatinized) starch plastic or from starch-filled low-density polyethylene (LDPE) or polypropylene. The study concluded that conventional moulded pulp fabrication is not a viable option to cost-effectively and sustainably meet the enormous market demand in China for take-out/delivery containers. This is due to restrictive fiber morphology requirements, limited sources of suitable fiber in China, and the high water, chemical, energy and pollution intensiveness of the pulping and manufacturing processes. The required pulping and manufacturing processes were identified as more energy, water and pollution intensive than other domestic manufacture of disposable food containers such as plant-starch plastics and ground lignocellulosic residues.

These concerns are also raised in Western reports. Mordor Intelligence (2019) discusses US government environmental regulation of the pulping industry (which is highly water, energy and pollution intensive compared with polystyrene production) as a significant constraint to market growth of moulded pulp. The pulp and paper industry is the 4th largest emitter of GHGs and dioxin environmental pollutants and ranked third for water demand in the USA. The industry releases around 212 M metric tonnes of industrial air and water pollutants per annum. Moulded pulp manufacture consumes 70–115% more energy, generates up to 31% more air pollution, and emits 323–248% more GHGs (Mordor Intelligence, 2019). This will lead to more efficient fabrication pathways such as minimal pulping and dry thermoforming.

4.4. PFAS replacement

Tests on take-out food packaging show moulded pulp can contain high levels of PFAS (Fassler, 2019; LaMotte, 2020; Strakova et al., 2021) with significant environmental (leaching, mammal and environmental toxicity) and public health implications (Grinvalski, 2019; Schaider et al., 2017). Most are difficult to identify using target compound assay. Some products contain over 60 times the levels stipulated by the Danish Veterinary and Food Regulatory Authority which is sufficient to disrupt endocrine system function (Strakova et al., 2021). These concerns are so significant that developing PFAS-free compositable take-out containers is a top priority and the pulp and paper and foodservice industries are adapting rapidly. Change is being driven by country e.g. Denmark (Danish Veterinary & Food Administration, 2020). Several US states now prohibit PFAS in food packaging (Hogue, 2021; SGS, 2021; Washington State Department of Ecology, 2021) with other North American jurisdictions following suit. In 2020 four chemical companies discontinued manufacture of 6:2 fluoro-telemer alcohol, a short chain PFAS commonly used in moulded pulp food packaging (Hogue, 2021).

While around 60 short chain ($\leq$6 C) FC additives for food contact are still permitted by FDA in domestic manufacture or imported products in North America (Begley et al., 2005) PFAS-free moulded pulp containers are available (Business Wire, 2020) and voluntarily adopted by some fast food chains (Fassler, 2020; Washington State Department of Ecology, 2021). PFAS and related compounds are now prohibited for packaging product certification for biodegradability through the international Biodegradable Products Institute (BPI, 2020). Understanding the identity and properties of proprietary PFAS-alternatives is still very limited and early bio-accumulation and toxicity issues have been identified in certain replacement compounds (Hogue, 2021; Zhang et al., 2021). The use of Teflon and other PFAS compounds to prevent casts sticking to moulds (DTSC, 2021) is also identified as a risk factor complicating the accreditation and acceptance of moulded pulp products as PFAS-free packaging alternatives (Washington State Department of Ecology, 2021).

4.5. Research priorities

From the above discussions priority areas for research and development of fiber-based food packaging include: (1) developing knowledge on the potential of non-wood fibers for market pulp; (2) pretreatments and pulping routes to improve sustainability of fiber extraction and moulded pulp manufacture; (3) improving knowledge of non-wood fiber morphological, chemical and physical properties; (4) understanding the pulping/(bleaching)/refining interactions; (5) exploring different fiber blends (including wood fibers and/or recycled fibers) for synergies and properties tailoring; (6) applications of novel, non-toxic additives and systems in moulded pulp manufacture; (7) developing techno- and circular-economies (including viable pathways for recycling and re-using of recovered fiber); and (8) carrying out comparative life cycle analyses for different food container packaging substrates.

5. Conclusions

Increasing restrictions on single-use plastics open up market opportunities for moulded pulp food containers and packaging with renewable, biodegradable waste fiber sourcing and adaptable manufacturing technology. This paper discussed technical evolution and regulatory aspects of moulded pulp disposable food container production, product properties, fiber options, performance additives and disposal pathways. High monetary and environmental costs of virgin wood fiber and lack of clear guidelines on the use of recycled fiber has transitioned the moulded pulp foodware industry to unbleached virgin fiber from crop wastes (currently largely bagasse) and other fast grown biomass such as bamboo. Manufacturing has shifted to the largest growing markets in Asia where plastics such as XPS are also being restricted and industrial food/fiber crops like sugar cane, bamboo, and oil palm are cultivated. The take-out food container requires different functionality to standard moulded pulp packaging including enhanced moisture and grease resistance, no or limited migration from substrate to food, good thermal insulation and design safety to prevent spillage.

Challenges including slow production speed and high energy consumption are being met with evolution of basic moulded pulp process such as transfer moulding, thermoforming and dry thermoforming of pulp and pre-fabricated papers. Significant advances have also been made in crop-waste pulping, non-toxic additives and custom fiber treatments, reduced water- and energy-intensive manufacturing processes, and rapid prototyping/3D printing of forming and pressing moulds. Biodegradable, non-toxic plant-derived additives and liners are now available for paper products and packaging applications, and further research is required to evaluate their effectiveness for performance enhancement compared to PFAS and commercialize these into functional but safely compostable/recyclable take-out containers.

Complex legislative and market signals create challenges for the much-needed transition to plant fibers and polymers for packaging to replace petroleum plastics. Plastics such as polypropylene remain competitive in the take-out container market for their cost, durability, uniformity, speed and ease of processing and clear, well-established pathways for recycling and re-using of recovered fiber. Bioplastics have evolved greatly and benefit from the efficient and mature technology of plastics manufacture. However, these are materially inefficient, face supply chain problems, and require nano-scale additives and/or co-addition of petroleum polymers for processability and functionality which hamper biodegradation and recycling, and create environmental contamination.

To retain market competitiveness biomass-based packaging requires improved compost/recycling waste pathways, further product development and proactive government policies and municipal waste management programs. More published data are needed on properties and relative advantages/disadvantages of different kinds of so-called ‘green’ take-out packages, comparing moulded pulp with others made from bioplastics. Significantly missing is a comprehensive LCA study of different kinds of moulded pulp container compared with their equivalents made from XPS, polypropylene, bioplastic or aluminium.

CRediT authorship contribution statement

Katherine E. Semple: Investigation, Writing – original draft, Writing – review & editing, Software. **Chenli Zhou:** Investigation, Writing – review & editing, Data curation, Software. **Orlando J. Rojas:** Writing – review & editing. **William Nguegang Nkeuwa:** Review and Software. **Chunping Dai:** Funding acquisition, Conceptualization, Supervision, Writing – review & editing, Project administration.

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